

# Accepted Service Adaptation: Tight Parametric Quotients and Minimal Additional Writable Memory

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**Abstract.** Public renewal states can recombine while carrying different accepted continuation obligations. For positive additive payments and separable quadratic rewards, the unfolded problem is classical laminar allocation, and normalized memoization yields the same public-graph recursion. We characterize the residual representation problem. An occurrence-to-vertex quotient constructs the complete dual-price response with at most  $3N$  global knots and  $O(N^2 + M)$  response storage; both orders are worst-case tight, and a coefficient-event implementation has polynomial rational bit complexity including exact sorting and comparison. For a fixed root promise, the reached price graph is a deterministic output transducer. Classical state minimization then gives the smallest equivalent reached machine, while contract structure implies ordered behavior classes, a minimum additional writable alphabet  $K_* = \max_v c_v$ , and a linear worst-case family. We separately report combined reached states and read-only response precision. A heterogeneous renewal model yields an exact memory-value frontier and a strict comparative static along every ordered mean-preserving radial spread of continuation caps. A graph-sized shared-table comparator and exact supporting cuts complete the service benchmark. Exact laminar, convex, and public-graph comparisons validate the representation and repeated-query use.

**Keywords:** continuation participation; contractual memory; parametric resource allocation.

**Subject classifications:** Programming: stochastic resource allocation; Dynamic programming/optimal control: contractual memory.

**Area of review:** Stochastic Models.

# 1 Introduction

A service agreement can return to the same public operating condition after different sequences of renewal reviews. Those histories need not permit the same future service. A customer who could have exited at an earlier review may have accepted a different remaining payment obligation. A provider that remembers only the current public condition can therefore lose value even when demand is Markovian, prices are fixed, and neither party has private information. The operational question is not whether full history can improve one schedule. It is whether the accepted optimum admits a compact exact representation, how large that representation must be in the worst case, and how much additional writable information exact execution requires.

We answer these questions for positive additive payments, separable strictly concave quadratic operating rewards, interval controls, and an exogenous recombining public graph. The exact compiler excludes intertemporal switching costs and coupled vector commitments. These conditions make one accumulated payment price sufficient for optimal continuation. Each participation cap reflects that price at a vertex-specific barrier. We construct the complete parameterized response family before choosing a root promise; the construction neither obtains an extensive-form optimum first nor enumerates a scenario tree.

The classical starting point is explicit. After probability weighting, the unfolded problem is separable concave resource allocation with laminar descendant constraints, in the lineage of Mjelde (1983) and Tang (1990). Normalized memoization of that occurrence-indexed reduction produces the same public-graph recurrence. Parametric solution paths and sensitivity are also established themes: Klimm and Warode (2021) give output-polynomial algorithms for piecewise-quadratic parametric flow, while Harks et al. (2018) study sensitivity of separable optimization over integral polymatroids. We therefore claim neither scalar marginal allocation, memoization, generic piecewise-affine parametricity, nor state minimization as new principles. The residual contribution is quantitative and representation-specific: on the compact public graph, the complete accepted-service response has at most  $3N$  distinct knots, this linear knot bound and the resulting quadratic response-storage order are both worst-case tight, and the coefficient-event implementation has polynomial rational bit complexity.

The second contribution concerns execution rather than compilation. The reached price graph is a deterministic output transducer driven by realized public transitions. Backward state minimization is classical for deterministic machines and related state-aggregation models (Bean et al., 1987; Givan et al., 2003; Peyrière, 2023). Applying that principle to the reached transducer yields the minimum equivalent behavior partition. The contract-specific statements are that a continuous-control problem generates a finite reached machine; its behavioral classes are contiguous in price order; the additional writable alphabet equals the largest within-vertex class count; distinct numerical prices can collapse to one behavior; and a renewal family attains the linear worst-case memory order. We report separately the additional writable alphabet, the number of reached public-state/class pairs, and the read-only response representation and coefficient precision.

The renewal family also gives an exact value–memory frontier with heterogeneous branch probabilities and operating costs. Along every strictly ordered mean-preserving radial spread of continuation caps, insufficient deterministic memory has strictly increasing loss, while the exact alphabet requirement can jump at arbitrarily small spread. The statement is a radial comparative static, not a claim of monotonicity under

every convex-order or majorization relation. The randomized exact-optimum lower bound uses an explicit information-accounting convention; pathwise feasibility with respect to private controller randomization only strengthens that lower bound.

The service interpretation is positioned against finite policy graphs in dynamic contracting. Zhang (2012) uses finite policy graphs to approximate continuation contracts in an infinite-horizon adverse-selection model. Our model has public information and a finite horizon, and it does not solve adverse selection. Its policy graph is instead an exact compiler-generated transducer for a fixed root query. This distinction, together with the public-graph promise-grid comparator, prevents a graph representation itself from carrying the novelty claim.

The computational study is organized around the residual theorem claims. An independently implemented occurrence-indexed laminar reduction checks exact equality; a public-graph promise method checks a competing compact state representation; dense response families expose output size; and repeated root inversions measure the use of a compiled parameterized path. Timing against a deliberately nonmemoized tree is retained only as an implementation comparison, not as evidence against the equivalent normalized memoized reduction or against specialized parametric algorithms.

The broader accepted-adaptation results on switching, comparator-adjusted rents, continuous states, and certified deployment remain available in the retained theory volumes with their statements and proofs unchanged. They are not used to enlarge the domain of the finite-price theorem. The paper's contribution is therefore a sharp exact theorem package for the stated subclass: a compact quotient, tight response complexity, explicit bit complexity, and a precisely accounted minimal writable implementation.

## 2 Service Agreements and Their Laminar Allocation Form

### 2.1 Public primitives and accepted continuations

Let  $G = (\mathcal{V}, \mathcal{E})$  be a finite, rooted, acyclic public graph. Vertices include date; all are reachable. Write  $N = |\mathcal{V}|$ ,  $M = |\mathcal{E}|$ , and let  $p_{vw} > 0$  be the exogenous probability on edge  $(v, w)$ , with probabilities summing to one at a nonterminal vertex. The discount factor is  $0 < \beta \leq 1$ . An occurrence  $h$  is a root-to-vertex history in the finite unfolding  $\mathcal{T}$ ; its public vertex is  $v(h)$  and its discounted reach weight is  $w_h = \beta^{t(h)} \Pr(h) > 0$ . Write  $j \succeq h$  for a descendant, including  $h$ , and  $H = |\mathcal{T}|$ .

At occurrence  $h$  of  $v$ , the provider chooses a tier  $x_h \in [l_v, h_v] \subseteq [0, 1]$ . The payment coefficient  $a_v > 0$  is fixed, and the operating reward is  $g_v(x) = r_v x - q_v x^2/2$ , with  $q_v > 0$ . Customer utility is a publicly specified service-benefit term minus  $a_v x$ . For example, fixed stock  $s_z$  yields

$$u(z, s_z, x) = v \mathbb{E} \min(s_z, D_z) - a_z x, \quad a_z = r_z^{\text{premium}} - \kappa \mathbb{E}(D_z - s_z)^+ > 0. \quad (1)$$

Stock may be stipulated, or identified by the service and cost commitments in the retained stock-identification results. The customer can leave before each modeled review; the provider commits to its contingent protocol. Retaining the customer requires the conditional service benefit minus remaining payment to exceed the public

outside continuation. Assume their difference is a vertex-sufficient cap  $B_v$ . The accepted optimization is

$$\begin{aligned} J(b_0) = \max_x \quad & \sum_{h \in \mathcal{T}} w_h g_{v(h)}(x_h) \\ \text{subject to} \quad & Y_h := \sum_{j \succeq h} \frac{w_j}{w_h} a_{v(j)} x_j \leq B_{v(h)} \quad (h \in \mathcal{T}), \quad Y_0 = b_0, \\ & l_{v(h)} \leq x_h \leq h_{v(h)}. \end{aligned} \quad (2)$$

The root has a cap as well as the equality promise. No privately informed reports, endogenous termination choice, or policy-dependent transition probabilities are part of this institution. The no-switching restriction is essential to the scalar-price result; the retained general model continues to allow switching and is not replaced by (2).

The feasible remaining-payment interval is computed without unfolding:

$$L_v = a_v l_v + \beta \sum_w p_{vw} L_w, \quad U_v = \min \left\{ B_v, a_v h_v + \beta \sum_w p_{vw} U_w \right\}. \quad (3)$$

A subtree is feasible precisely when its successors are feasible and  $L_v \leq U_v$ . Throughout,  $b_0 \in [L_0, U_0] \subseteq (-\infty, B_0]$ . Convexity fills every point between the two extreme payments. Compactness gives an optimum and strict concavity gives a unique occurrence-indexed action vector.

## 2.2 An exact mapping, not an analogy

**Proposition 2.1** (Laminar allocation equivalence). *Define  $z_h = w_h a_{v(h)} x_h$ . Problem (2) is exactly the separable concave allocation problem*

$$\begin{aligned} \max_z \quad & \sum_h \left[ \frac{r_{v(h)}}{a_{v(h)}} z_h - \frac{q_{v(h)}}{2w_h a_{v(h)}^2} z_h^2 \right], \\ & \sum_{j \succeq h} z_j \leq w_h B_{v(h)}, \quad \sum_j z_j = b_0, \\ & w_h a_{v(h)} l_{v(h)} \leq z_h \leq w_h a_{v(h)} h_{v(h)}. \end{aligned} \quad (4)$$

*The constrained descendant sets are laminar. The change of variables is invertible and preserves objective values and optimal policies.*

*Proof.* Substitution gives each term and row of (4). Two rooted descendant sets in a tree are either disjoint or one contains the other. Positivity of  $w_h a_{v(h)}$  makes the coordinate map bijective and preserves every inequality direction.  $\square$

Thus the expanded primal lies in the tree-constrained allocation lineage of Mjelde (1983) and Tang (1990). Tang reports a reduction to one-variable convex subproblems; that result is part of the inherited optimization structure, not a competitor we relabel. The chain-nested allocation results of Vidal et al. (2016);

Schoot Uiterkamp et al. (2021); Wu et al. (2021) concern related but different input structures. A complexity bound for an ascending chain cannot simply be assigned to arbitrary branching descendant sets.

After subtracting lower allocations, the packing region is a laminar polymatroid, and the exact total selects a base of its truncation (Electronic Companion, Section EC.1). This connects the model to Federgruen and Groenevelt (1986) and Groenevelt (1991). Harks et al. (2018) further show that sensitivity analysis for separable optimization over integral polymatroid bases is a substantial literature in its own right. Their decision set is integral and their perturbations concern costs and polymatroid parameters; our theorem concerns a continuous quadratic accepted-service model and a complete response in one dual price. We cite that work as sensitivity lineage, not as a result to be rediscovered.

### 2.3 What remains after memoization and generic parametric optimization

An occurrence-indexed marginal reduction may represent subtree payment in the unnormalized units  $z$ . Its entire response is the occurrence weight times a common conditional response, proved below. Dividing by that weight makes occurrences of the same public vertex identical. Memoizing those normalized responses therefore gives our graph recursion. There is no additional stochastic obstruction or different first-order condition. We do not assert that scalar marginal allocation, subtree reduction, or memoization originated here.

The closest modern parametric comparison is Klimm and Warde (2021), who construct output-polynomial algorithms for separable, continuous, piecewise-quadratic parametric minimum-cost flow. Their parameter scales network demands and their feasible set is a flow polyhedron; our parameter is the price conjugate to an accepted-service payment and the unfolded feasible set is laminar allocation. A laminar problem may itself be reformulated before a parametric method is applied, so we make no priority claim for generic piecewise-affine solution paths or for output-sensitive parametric optimization. The contribution claimed here is narrower and quantitative: after quotienting repeated occurrences to the supplied public graph, all response changes are generated by local clipping events and at most one cap barrier per public vertex. This gives an at-most- $3N$  global knot universe, a matching  $\Omega(N)$  family, a matching  $\Omega(N^2)$  total-response family, and an explicit rational bit bound measured in the compact graph input. These conclusions are proved directly in Section 3.

This positioning answers five distinct questions. First, the complete response is a special structured parametric convex solution map, so pathwise parametricity itself is not claimed. Second, an unfolded laminar algorithm can compute the same path, and normalized memoization yields the same recurrence. Third, the compact quotient makes the response-size bound depend on public vertices rather than occurrence histories. Fourth, the global knot and total-storage orders are shown below to be tight for polynomial-size rational instances. Fifth, the bit bound is for the explicit coefficient-event constructor and includes exact sorting and comparison cost; it is not a generic encoding theorem for every parametric solver.

## 2.4 Behavioral minimization and policy-graph lineage

The reached execution object is also connected to classical minimization. Bean et al. (1987) study aggregation in dynamic programming; Givan et al. (2003) develop state equivalence and model minimization for MDPs; and Peyrière (2023) gives a minimum-state equivalent Moore-machine construction. Our backward equality of current outputs and successor classes is an instance of that classical deterministic behavioral-minimization principle. Section 4 therefore states the reached price system as an output transducer first, applies standard minimization, and then isolates the contract-specific results: finite reachability from continuous controls, ordered contiguous behavior classes, separate writable/read-only accounting, and a linear contractual lower-bound family.

Promised continuation utility is a standard recursive state in dynamic contracting (Spear and Srivastava, 1987; Thomas and Worrall, 1988; Chen et al., 2020). Zhang (2012) is especially relevant because it uses finite policy graphs to represent continuation contracts and payoff frontiers in an infinite-horizon adverse-selection model with Markov information. That policy graph is an approximation device for a different information and incentive problem. Our public-information model is finite horizon, has no private reports, and compiles an exact transducer for a fixed root promise. The comparison is nevertheless conceptually close: the present result can be read as an exact policy-graph compression theorem for this scalar public-information subclass.

Policy-graph decomposition and stochastic decomposition already avoid naive scenario enumeration (Pereira and Pinto, 1991; Girardeau et al., 2015; Dowson, 2020). We compare an explicit primal promise-grid method on the public graph rather than treat the mere use of a graph as a contribution. Our exact closure depends on one scalar additive commitment and separable quadratics; general convex policy-graph decomposition has a broader domain. The supporting cuts for fixed tables are also instances of standard dual decomposition (Benders, 1962; Boyd and Vandenberghe, 2004), not a new general finite-convergence theorem.

**Input-size convention.** An explicit-tree representation has  $H$  local records and must at least expose those records to an occurrence-indexed implementation. Our compact representation has  $N$  vertices and  $M$  edges. This is an input/output comparison, not a lower bound against every algorithm that accepts a compact graph. The independent tree code is retained as an exact cross-check, not as a stand-in for the strongest classical or modern specialized solver.

## 3 A Tight Quantitative Quotient of the History Tree

### 3.1 Conditional responses and cap reflection

For an occurrence  $h$  of  $v$ , let  $H_h(\eta)$  be the maximum conditional reward minus  $\eta Y_h$ , imposing all subtree caps but not an exact payment equality at  $h$ . Let  $H_h^{\text{pre}}(\eta)$  remove only the cap at  $h$ , retaining every successor cap. Conditional rewards use weights  $w_j/w_h$ . Define  $R_h(\eta)$  as the payment of the unique maximizer of  $H_h(\eta)$ ; use  $d_h(\eta)$  for the pre-cap payment. These definitions separate a removed current cap from the constrained continuation problems below it.

**Theorem 3.1** (Occurrence-to-vertex quotient and finite response closure). *Under the assumptions of Section 2, occurrences  $h, h'$  of a public vertex  $v$  with the same incoming price  $\eta$  have identical normalized optimal continuation plans. In particular,  $H_h = H_{h'} =: H_v$ ,  $H_h^{\text{pre}} = H_{h'}^{\text{pre}} =: H_v^{\text{pre}}$ , and  $R_h = R_{h'} =: R_v$ . The following backward construction computes the common payment responses and optimal policies:*

$$x_v^0(\eta) = [(r_v - a_v\eta)/q_v]_{[l_v, h_v]}, \quad (5)$$

$$d_v(\eta) = a_v x_v^0(\eta) + \beta \sum_w p_{vw} R_w(\eta), \quad (6)$$

$$R_v(\eta) = \min\{B_v, d_v(\eta)\}. \quad (7)$$

Write  $d_v^- = a_v h_v + \beta \sum_w p_{vw} U_w$  for the finite constant on the left tail of  $d_v$ . If  $B_v \geq d_v^-$ , assign no barrier at  $v$ . Otherwise let  $\alpha_v$  be the leftmost finite solution of  $d_v(\alpha_v) = B_v$ . The optimal local action and the incoming price sent to every successor are determined by

$$s_v = \max\{\eta, \alpha_v\}, \quad x_v = x_v^0(s_v), \quad (8)$$

where a missing barrier leaves  $s_v = \eta$ .

Every  $R_v$  is continuous, nonincreasing, piecewise affine with constant tails and range  $[L_v, U_v]$ . The global universe of distinct response knots has size at most  $3N$ . Complete response tables can be constructed in

$$A = O((N + M)N \log(N + 1)) \quad (9)$$

exact rational arithmetic operations and exact comparisons, using  $O(N^2 + M)$  scalar storage. For a fixed root promise, choose any finite  $\eta_0$  with  $R_0(\eta_0) = b_0$ . Execution uses at most  $N + 1$  distinct incoming-price labels, drawn from  $\eta_0$  and the finite cap barriers. The construction does not enumerate  $H$  occurrences.

*Proof.* The subtree below an occurrence is determined by its displayed vertex and subsequent public transitions. The map replacing the initial occurrence by another occurrence of the same vertex preserves all suffix paths, relative probabilities, discounts, local rewards, bounds, and caps. It is consequently a bijection of the normalized feasible problems at any common  $\eta$ . Strict concavity makes their maximizing action vectors agree. This proves the quotient independently of the algorithm.

Remove the current cap. Separability gives (5) and independent successor problems at the same incoming price, proving (6). Their unique maximizers have continuous payment responses. Backward induction starts with a clipped affine local payment and preserves continuity, monotonicity, and affine pieces under weighted summation. Its two tails are the extreme attainable payments before the current cap.

When the current cap is redundant, this pre-cap maximizer is feasible. Otherwise feasibility implies that  $B_v$  belongs to the range of  $d_v$ . The clipped finite response reaches its tails at finite prices, so a finite leftmost crossing exists. For  $\eta < \alpha_v$ , the pre-cap optimum at  $\alpha_v$  has payment  $B_v$ . Let  $z^*$  be that optimum and  $z$  any policy satisfying the current cap. Writing  $G$  for conditional reward and  $Y$  for payment gives

$$G(z) - \eta Y(z) \leq G(z^*) - \alpha_v B_v + (\alpha_v - \eta) Y(z) \leq G(z^*) - \eta B_v.$$

For  $\eta \geq \alpha_v$ , the pre-cap optimizer already satisfies the cap. These two cases prove reflection, (7), and its implementing policy. If the payment is constant between two prices, the two optimality inequalities imply that either unique optimizer is also optimal at the other price. Thus the policy is unchanged on an equality plateau even when the supporting dual price is not unique. This also covers a cap at the minimum feasible payment and fixed local intervals.

Local clipping contributes at most two knots. Summing child responses introduces no knot outside their union. Capping can remove knots and introduces at most one crossing. Inducting over distinct *vertices*, rather than occurrences, gives a global union of at most  $3N$ . Each of the  $N$  responses can still have order  $N$  segments, so total response storage is quadratic, not linear. Directly merging weighted coefficient changes at sorted knots gives (9); the graph takes an additional  $O(N + M)$  records.

Finally, each executed price is the maximum of the root price and the barriers along the realized path. There are at most  $N + 1$  such numerical labels for this fixed root query, and at most  $N(N + 1)$  reachable vertex–price pairs. The root maximizer at  $\eta_0$  has precisely  $b_0$ , so it also maximizes reward among policies satisfying the equality.  $\square$

The unnormalized response at occurrence  $h$  in Proposition 2.1 is  $w_h R_{v(h)}(\eta)$ . This identity is the explicit link to an occurrence-indexed tree reduction. A normalized memoized implementation enjoys the same quotient recurrence; it is not an algorithmic competitor that our implementation is claimed to dominate.

### 3.2 Matching lower bounds for the response representation

The preceding upper bounds are not artifacts of a loose counting argument.

**Theorem 3.2** (Worst-case tight response size). *For every integer  $n \geq 1$ , there is an  $n$ -vertex rational instance with  $M = n - 1$  for which the global response-knot universe has exactly  $2n$  distinct knots and the total number of affine response segments is exactly  $n^2 + 2n$ . Consequently the  $O(N)$  global-knot bound and the  $O(N^2 + M)$  complete-response storage bound in Theorem 3.1 are worst-case tight up to constant factors, even with no binding continuation caps.*

*Proof.* Take the deterministic chain  $1 \rightarrow 2 \rightarrow \dots \rightarrow n$ , with  $\beta = 1$ , edge probability one,  $a_i = q_i = 1$ ,  $l_i = 0$ ,  $h_i = 1$ , and  $r_i = 2i$ . Set the continuation cap at vertex  $i$  to

$$B_i = n - i + 1.$$

This cap equals the maximum feasible remaining payment, so it is redundant and creates no barrier. The response at vertex  $i$  is therefore

$$R_i(\eta) = \sum_{j=i}^n [2j - \eta]_{[0,1]}. \quad (10)$$

For every  $j$ , the local summand has knots  $2j - 1$  and  $2j$ . These  $2n$  numbers are distinct. At each such number the slope of exactly one summand changes by one in magnitude, so no knot cancels in any ancestor response that contains that summand. Hence  $R_i$  has exactly  $2(n - i + 1)$  knots and  $2(n - i + 1) + 1$  affine

segments. Summing over  $i$  gives

$$\sum_{i=1}^n [2(n-i+1) + 1] = \sum_{k=1}^n (2k+1) = n^2 + 2n.$$

All primitives are integers with  $O(\log n)$  bits, so the family has polynomial encoding length. The claimed asymptotic lower bounds follow.  $\square$

The theorem separates two notions that can otherwise be conflated. There are only linearly many distinct knot *locations*, but a downstream knot can appear in the response table of every ancestor, producing a quadratic number of stored affine pieces. The dense computational family in Section 6 is an empirical counterpart of this analytic construction; the lower bound itself does not depend on computation.

**Corollary 3.3** (Several coordinates under one scalar commitment). *With  $d_v$  separable local coordinates, positive payment coefficients, positive quadratic coefficients, and a product interval, let  $D = \sum_v d_v$ . The knot universe has size at most  $2D + N$ , response storage is  $O(N(2D + N) + M)$ , and the incoming-price alphabet still has at most  $N + 1$  elements for a fixed query.*

*Proof.* There are two clipping knots per coordinate, but only one scalar cap barrier per vertex. Clipping knots describe how a local action responds to a given price; they are not additional prices written into the running record. The common commitment price and cap reflection are unchanged.  $\square$

Independent commitments can be separated only when rewards and feasible sets also separate. Coupled vector promises, nonseparable quadratic terms, and intertemporal switching do not satisfy the scalar reflection argument. Their retained primal promise-state and convex formulations remain applicable.

### 3.3 Rational bit complexity

An arithmetic count alone does not establish polynomial time on binary inputs. Let  $L$  be the total binary encoding length of all rational primitives, graph indices, and the rational root query, with rationals in lowest terms. The following statement is specifically for the coefficient-event construction used in Theorem 3.1. Stored rational coefficients, event locations, and crossings are reduced to lowest terms after each exact arithmetic update. The common denominators used in the proof below are bounding devices; they are not materialized as a giant implementation denominator.

**Theorem 3.4** (Polynomial rational preprocessing for the coefficient-event compiler). *All stored response slopes, intercepts, finite knots, cap barriers, and a returned finite root price can be represented with*

$$S = O(NL + \log(N + M + 1)) \tag{11}$$

*bits per reduced numerator and denominator. The event merges, sorting, exact comparisons, coefficient arithmetic, cap-crossing calculations, and one root inversion together admit the conservative bit-operation bound*

$$O((N + M)N \log(N + 1) S^3) = O(AS^3). \tag{12}$$

*The read-only response representation occupies  $O((N^2 + M)S)$  bits. These are polynomial bounds for rational inputs to this explicit compiler, not a strongly polynomial claim and not a generic bit bound for every parametric optimization algorithm.*

*Proof.* Fix an open interval between global knots. At a vertex whose current cap binds on that interval, the payment response is the constant  $B_v$ . Otherwise its coefficients are a local affine payment plus the discounted, probability-weighted coefficients at successors. The possible nonzero local sources are

$$B_v, \quad a_v l_v, \quad a_v h_v, \quad a_v r_v / q_v, \quad -a_v^2 / q_v. \quad (13)$$

Expanding either coefficient through the acyclic graph therefore gives a sum over directed paths of one source multiplied by edge probabilities and a power of  $\beta$ . Expansion stops at a capped vertex. No path repeats an edge, and its length is at most  $N - 1$ .

Let  $D_0$  be the product of all positive denominators and the absolute values of all nonzero numerators appearing in the rational primitive input. Its logarithm is  $O(L)$ . Every path-term denominator divides  $D_0^{N+4}$ ; this exponent covers repeated discounting, each edge probability, and the products and divisions in (13). Thus a conceptual common denominator has  $O(NL)$  bits even though the number of paths may be exponential. The sum of nonnegative discounted path weights to source occurrences is at most  $N$ , because it is bounded by the expected number of visited vertices. The absolute value of a source is at most  $2^{O(L)}$ . Multiplying by the conceptual denominator therefore bounds every reduced stored numerator and denominator by (11).

The actual event compiler does not build  $D_0^{N+4}$ . It adds and multiplies ordinary reduced rationals while merging sorted coefficient events. Intermediate stored partial sums have the same path-product denominator structure and differ by at most a polynomial factor in  $N + M$  in magnitude. A new cap crossing has form  $(B_v - c)/m$  on a nonconstant line  $m\eta + c$ ; local clipping knots and a root inverse are ratios of a constant number of bounded-height rationals. Their reduced heights also satisfy (11).

Sorting and merging are part of the operation count in (9). Every exact order test is a cross-product comparison of  $O(S)$ -bit integers. Rational addition, multiplication, division, comparison, and gcd reduction can all be bounded conservatively by  $O(S^3)$  bit operations. Multiplying that cost by the full arithmetic-and-comparison count proves (12). Counting stored coefficients, knot locations, and graph entries proves the storage claim.  $\square$

The coefficient height above is not the bit length of a completely printed primal–dual certificate. Aggregate execution rewards and certificate totals can combine different realized denominators and have a larger, still polynomial, height. The Electronic Companion reports that separate aggregate bound and the implementation records coefficient precision separately from certificate precision. This distinction is part of the memory accounting below.

**Corollary 3.5** (Amortized exact root inversion). *After the complete response family is compiled, a new feasible root promise can be inverted on the stored root response using  $O(\log(N + 1))$  exact comparisons and a constant number of rational arithmetic operations. This is an inversion bound only: emitting and auditing the reached execution machine for the new query is a separate operation.*

*Proof.* The root response has at most  $3N$  globally indexed knots. Binary search locates the constant or affine segment containing the requested payment; one affine solve, or selection of a finite point on a constant tail, returns a valid root price.  $\square$

### 3.4 A compact exact certificate

For each reached  $(v, \eta)$ , propagate discounted occupation weight  $w$ , effective price  $s$ , action  $x$ , and multipliers  $\chi = s - \eta$ ,  $\ell, u \geq 0$ . They satisfy

$$r_v - q_v x - a_v s + \ell - u = 0, \quad \ell(x - l_v) = u(h_v - x) = \chi(B_v - R_v(\eta)) = 0. \quad (14)$$

Weight  $\beta w p_{vw}$  moves to  $(w, s)$ . A backward conditional-payment calculation verifies every cap; forward flow verifies occupation weights and the root promise. The resulting dual upper bound is

$$\eta_0 b_0 + \sum_{(v, \eta)} w \left[ \chi B_v + u h_v - \ell l_v + \frac{(r_v - a_v s + \ell - u)^2}{2q_v} \right]. \quad (15)$$

Ancestor prices telescope under the flow equations, and (14) makes (15) equal the implemented reward. These rational checks certify a particular optimum; verifying entire response-curve identities is a separate implementation test. Neither kind of code execution replaces the mathematical proof above.

## 4 Minimal Additional Writable Memory and Its Contractual Value

### 4.1 Observation architecture and the reached transducer

Fix the public graph, all primitives, and one root promise. At a decision, the decoder observes the current public vertex, including date, and a writable symbol. A read-only table may contain primitives, response coefficients, thresholds, the query's root price, and transition/output maps. Between decisions the update may depend on the old symbol and the realized public transition; the next action decoder has no separate access to the past transition. A retained promise, past tier, branch identity, or random seed must be charged as writable information if it carries history not contained in the current public vertex. A continuous remaining promise is therefore not an uncharged alternative register.

This architecture deliberately separates three resources. The *additional writable alphabet* is the changing record carried across recombination. The *combined reached behavior states* are the observable public vertex paired with a behavior class. The *read-only representation* contains the graph, compiled responses, coefficient precision, and transition/output tables. None of these quantities is used as an unqualified notion of memory below. Changing the root promise requires a new inversion and can change the reached state set.

Let  $\mathcal{P}_v$  be the incoming prices reachable at  $v$  under Theorem 3.1. For  $\eta \in \mathcal{P}_v$ , put  $s_v(\eta) = \max\{\eta, \alpha_v\}$  with the missing-barrier convention. The reached system is a deterministic output transducer. Its state set is

$$Q = \{(v, \eta) : v \in \mathcal{V}, \eta \in \mathcal{P}_v\}.$$

At state  $(v, \eta)$  the output is  $x_v^0(s_v(\eta))$ . If the realized public transition is edge  $(v, w)$ , the next transducer state is  $(w, s_v(\eta))$ . The public edge is thus an exogenous input to the deterministic machine.

For two prices at the same public vertex, define behavioral equivalence backward:

$$\eta \sim_v \eta' \iff \begin{cases} x_v^0(s_v(\eta)) = x_v^0(s_v(\eta')), \\ s_v(\eta) \sim_w s_v(\eta') \text{ for every } (v, w) \in \mathcal{E}. \end{cases} \quad (16)$$

At a terminal vertex only the first condition is needed. This is the usual equality-of-output-and-successor-class recursion used in deterministic machine minimization. We use that classical principle explicitly (Bean et al., 1987; Givan et al., 2003; Peyrière, 2023); the new content lies in the contract-generated transducer and its ordered structure.

**Theorem 4.1** (Minimal reached transducer and additional writable alphabet). *Let  $c_v$  be the number of classes of  $\sim_v$  on  $\mathcal{P}_v$ . Under the preceding observation architecture:*

1. *the reached controller has*

$$C_* = \sum_{v \in \mathcal{V}} c_v \quad (17)$$

*reachable public-state/class pairs;*

2. *the minimum size of a common additional writable alphabet is*

$$K_* = \max_{v \in \mathcal{V}} c_v, \quad (18)$$

*hence the changing record needs  $\lceil \log_2 K_* \rceil$  bits;*

3. *every class is contiguous in the ordered list of incoming prices at its public vertex;*
4. *the backward signatures can be constructed on the reached vertex-price graph in  $O((N+M)N \log(N+1))$  arithmetic comparisons and elementary index operations.*

*The read-only response compiler remains charged separately: it stores  $O(N^2 + M)$  rational scalars of coefficient height at most  $S$  from Theorem 3.4. A direct class-transition table has  $\sum_v c_v \deg^+(v)$  entries. The exact-optimum lower bound for  $K_*$  also holds for randomized controllers whose retained private randomness is charged to the same writable state and whose continuation feasibility is imposed in conditional expectation at each public history.*

*Proof.* For a finite deterministic output transducer, two states can share one minimized state exactly when they emit the same current output and every common input leads to equivalent successors. Applied within a fixed observable public vertex, this is precisely (16). Backward partition refinement therefore yields the coarsest behavior-preserving partition of the reached price states. This is the classical minimization step.

Under the declared architecture the public vertex is observed without using the writable register. Thus each pair  $(v, \text{class})$  is a distinct reached combined state, giving (17), while class indices may be reused across

different public vertices. A common alphabet of size  $\max_v c_v$  is sufficient: at vertex  $v$ , assign distinct labels only to its  $c_v$  classes, decode the common action, and update to the corresponding child label. Conversely, if two inequivalent prices at the same public vertex share one writable symbol, the deterministic decoder and update maps coincide from that point onward, so some positive-probability suffix receives a different action than the unique optimum. Hence at least  $c_v$  labels are necessary at each  $v$ , proving (18).

Along a fixed suffix path, the effective price is the maximum of the incoming price and a fixed finite set of barriers. Every local action is nonincreasing in that effective price. Thus every coordinate of the complete future plan is nonincreasing in the initial incoming price. If two ordered endpoint prices induce the same plan, every intervening price induces that same plan coordinate by coordinate. Hence each equivalence class is contiguous.

A signature consists of the local action and the tuple of successor-class indices. Processing signatures in reverse topological order over at most  $N(N + 1)$  reached vertex–price pairs and at most  $M(N + 1)$  pair transitions gives the stated conservative bound. The read-only counts are direct consequences of Theorems 3.1 and 3.4.

For a randomized feasible controller, take the mean action at each public history. Exogenous transitions and linear payments and caps make this mean vector feasible under conditional-expectation feasibility. Strict concavity gives

$$\sum_h w_h \mathbb{E} g_{v(h)}(X_h) \leq \sum_h w_h g_{v(h)}(\mathbb{E} X_h) \leq J(b_0),$$

with equality only if each action is almost surely constant at the unique optimal value. Distinct required continuation plans must therefore have disjoint supports of retained symbols; a shared symbol with positive probability would impose the same future behavior on both histories. Additional retained randomness capable of distinguishing them is itself writable information under the stipulated accounting. At least  $c_v$  symbols are necessary.  $\square$

The randomization convention has a concrete contractual interpretation. Conditional-expectation feasibility applies when participation is evaluated before an internal controller coin is realized and the private coin is not a contractually observable service state. If the institution instead requires every private randomization realization to satisfy every continuation cap pathwise, the admissible controller class is smaller. The deterministic optimum and the lower bound above remain valid; no smaller exact alphabet can be created by imposing the stronger feasibility requirement.

**Corollary 4.2** (Linear writable-memory order is tight). *For every  $k \geq 1$ , the renewal family in Theorem 4.3 has  $N = k + 2$  public vertices and satisfies  $K_* = k = N - 2$ . Hence the  $O(N)$  upper bound on the additional writable alphabet is worst-case tight.*

*Proof.* The  $k$  intermediate branches recombine into one terminal public vertex and require  $k$  distinct optimal terminal actions. They are therefore  $k$  distinct behavioral classes at that terminal vertex. The root and terminal add two public vertices.  $\square$

The theorem minimizes exact execution memory for this fixed optimum under a declared observation interface. It does not minimize the response compiler’s storage or characterize the best approximate controller

for every alphabet size. Those are separate design questions. The distinction is now explicit in  $K_*$ ,  $C_*$ , and the read-only response size.

## 4.2 Ordered memory frontiers beyond equal branch probabilities

Consider three dates. The root tier is fixed at zero. At the intermediate date a branch  $j$  is observed with probability  $\pi_j > 0$ , where  $\sum_j \pi_j = 1$ . All branches then enter the same terminal public vertex. Payments and discounting equal one. Intermediate and terminal tiers are in  $[0, 1]$ . Branch  $j$  has cap  $0 < b_1 < \dots < b_k < 1$ , and the root promise is  $\bar{b} = \sum_j \pi_j b_j$ . At the intermediate date the reward is  $-h_j(y)$ , where  $h_j$  is convex, nondecreasing, and  $h_j(0) = 0$ . The common terminal reward  $f(c)$  is continuous, concave, and strictly increasing on  $[0, 1]$ . The terminal cap is one. The terminal decoder observes only its public vertex and its memory symbol, not the inherited intermediate tier.

**Theorem 4.3** (Heterogeneous renewal frontier). *The unique full-information policy is  $y_j = 0$ ,  $c_j = b_j$ , with value  $J = \sum_j \pi_j f(b_j)$ . Exact implementation requires  $k$  symbols. With at most  $m$  deterministic terminal symbols, an optimal encoding has contiguous nonempty cells in cap order. For a cell  $i, \dots, j$ , its terminal tier is  $b_i$  and its loss is*

$$C(i, j) = \sum_{h=i}^j \pi_h [f(b_h) - f(b_i) + h_h(b_h - b_i)]. \quad (19)$$

The exact minimum loss is  $D_m = D(\min(m, k), k)$ , where

$$D(s, j) = \min_{s-1 \leq i < j} \{D(s-1, i) + C(i+1, j)\}, \quad D(0, 0) = 0, \quad (20)$$

and all other infeasible entries equal  $+\infty$ . With a precomputed cell-cost table this requires  $O(mk^2)$  arithmetic operations. For quadratic  $f$  and quadratic  $h_j$ , prefix sums of the probability-weighted coefficients compute each cell cost in constant arithmetic time.

*Proof.* The root promise is the mean of the branch caps, so every branch cap must bind. Hence  $y_j = b_j - c_j$  and  $0 \leq c_j \leq b_j$ . The branch reward  $f(c) - h_j(b_j - c)$  is strictly increasing on that interval. Full information therefore uniquely selects  $c_j = b_j$ ; different terminal actions require different symbols.

Within a memory cell all terminal actions coincide and cannot exceed its smallest cap. Monotonicity selects that cap and subtraction from the full-information reward gives (19). Fix any collection of cell minima. Every branch strictly prefers the largest eligible selected minimum, irrespective of its probability and intermediate cost. Reassigning branches in this way keeps each selected minimum in its own nonempty cell and produces contiguous cells without reducing value. Splitting at the final cell gives (20). Expanding the quadratic terms gives the stated prefix-sum implementation.  $\square$

This strengthens and includes the quadratic equal-probability frontier. With  $f(c) = 2c - c^2/2$ ,  $h_j(y) = y^2/2$ , and  $\pi_j = 1/k$ , (19) becomes

$$C(i, j) = \frac{2 - b_i}{k} \sum_{h=i}^j (b_h - b_i). \quad (21)$$

For  $k = 8$  and  $b_j = j/9$ , one, two, and three symbols lose 119/162, 5/18, and 49/324, respectively. Two writable bits permit four symbols, not three. The segmentation recurrence is standard; the contract-specific cell loss, ordered-cell sufficiency, and exact memory interpretation are the substantive statements.

The earlier nonsaturated two-branch example also remains useful. With branch caps 1/4 and 1, root promise 1/2, and the same quadratic rewards, the full optimum uses terminal tiers 1/4 and 3/4 and has value 27/32. An optimized public-only table uses common terminal tier 1/4 and intermediate tiers 0, 1/2, giving 13/32. The adaptation gain is 7/16 relative to the optimized restriction. At the full optimum both inherited intermediate tiers are zero, so making that tier observable does not evade this example's exact-memory lower bound. By contrast, the entire restricted-memory frontier above depends on not allowing a branch-specific intermediate tier to serve as an uncharged code.

### 4.3 Mean-preserving radial spreads of outside options

Fix positive probabilities, an interior mean  $\bar{b}$ , and strictly increasing numbers  $d_j$  with  $\sum_j \pi_j d_j = 0$ . Let

$$b_j(\varepsilon) = \bar{b} + \varepsilon d_j, \quad 0 < b_j(\varepsilon) < 1. \quad (22)$$

Keep the common service-benefit continuation fixed. The outside options are that continuation minus  $b_j(\varepsilon)$ . Increasing  $\varepsilon$  is therefore a mean-preserving radial spread of continuation caps and outside options along the fixed ordered direction  $d$ ; physical public transitions are unchanged.

**Proposition 4.4** (Mean-preserving radial spreads increase the value of contractual memory). *For the quadratic renewal family in (21), with arbitrary positive branch probabilities and every strictly ordered zero-mean direction  $d$ , every insufficient deterministic alphabet  $m < k$  has  $D_m(\varepsilon) > 0$  and  $D_m(\varepsilon)$  is strictly increasing in  $\varepsilon > 0$  throughout the admissible range. Moreover  $D_m(\varepsilon) = O(\varepsilon)$  as  $\varepsilon \downarrow 0$ . Exact execution needs one symbol at  $\varepsilon = 0$  and  $k$  symbols at every  $\varepsilon > 0$ .*

*Proof.* For a fixed contiguous partition with cell minimum  $i$ , the loss on branch  $h$  in that cell is

$$\pi_h \varepsilon (d_h - d_i) (2 - \bar{b} - \varepsilon d_i).$$

For  $h > i$ , its derivative is

$$\pi_h (d_h - d_i) (2 - \bar{b} - 2\varepsilon d_i) > 0.$$

If  $d_i \leq 0$  this is immediate. If  $d_i > 0$ , admissibility gives  $\varepsilon d_i < 1 - \bar{b}$ , so the last factor exceeds  $\bar{b} > 0$ . A partition with fewer than  $k$  cells has at least one positive term. Every candidate partition cost is therefore strictly increasing and positive. The minimum of finitely many strictly increasing candidate costs is strictly increasing: evaluate at the minimizing partition for the larger parameter. Each cost is  $O(\varepsilon)$  near zero. Distinct optimal terminal tiers give the alphabet statement.  $\square$

The proposition is intentionally a theorem about every ordered radial spread direction, not a claim about arbitrary convex-order, majorization, or variance comparisons between unrelated cap vectors. It is nevertheless stronger than an equal-weight or one-calibration statement: branch probabilities are arbitrary

positive numbers and the direction  $d$  is arbitrary subject to strict order and zero weighted mean. The economic distinction remains sharp: the exact alphabet can jump discontinuously at zero spread while the value loss from insufficient memory converges continuously to zero.

## 5 The Optimized Shared-Table Comparator

A prescribed observation restriction is optimized before its adaptation value is assessed. Let  $c(v) \in \{1, \dots, C_{\text{info}}\}$  be its information cell and let  $u_c$  be a single table chosen before uncertainty. The restricted value is denoted  $K^{\text{tab}}$ , keeping the number of cells distinct from the value.

**Proposition 5.1** (Zero-release graph formulation). *The optimized zero-release comparator is the convex quadratic program*

$$\begin{aligned} K^{\text{tab}} &= \max_{u, y} \sum_v \bar{w}_v g_v(u_{c(v)}) \\ y_v &= a_v u_{c(v)} + \beta \sum_w p_{vw} y_w, \quad y_v \leq B_v, \quad y_0 = b_0, \\ l_v &\leq u_{c(v)} \leq h_v. \end{aligned} \tag{23}$$

Here  $\bar{w}_v = \beta^{t(v)} \Pr(v)$  is the graph's forward reach weight. There are  $C_{\text{info}} + N$  variables and  $O(N + M + C_{\text{info}})$  coefficient occurrences. For a fixed shared table, an incoming absolute switching cost can also be represented exactly by its root term and one reach-weighted epigraph term per public edge.

*Proof.* A common table fixes every local action by the public vertex. Backward conditional expectation gives the displayed  $y_v$  at every occurrence of  $v$ . Thus all history caps are equivalent to the  $N$  displayed caps. Conversely, every feasible table and payment vector implements accepted actions at all histories. Summing rewards by public reach weights proves the objective identity. An edge uses two fixed table entries, so its switching cost is also determined without additional history.  $\square$

This supporting formulation uses standard convex optimization; it is not the central algorithmic novelty. In particular, a public table is not the full-information policy class. The difference  $J(b_0) - K^{\text{tab}}$  compares two optimized protocols on identical primitives.

At a fixed table and release  $\delta \geq 0$ , permit  $|x_h - u_{c(v(h))}| \leq \delta \omega_{c(v(h))}$ , where  $\omega_c \geq 0$ . Set

$$l_v^{\text{eff}} = \max(l_v, u_{c(v)} - \delta \omega_{c(v)}), \quad h_v^{\text{eff}} = \min(h_v, u_{c(v)} + \delta \omega_{c(v)}).$$

Empty intervals or an infeasible root promise are rejected. Otherwise the finite-price compiler solves this fixed query exactly.

**Proposition 5.2** (Supporting cuts including coincident bounds). *At every feasible fixed query  $(u, \delta)$ , physical and corridor multipliers can be split so that nonnegative corridor multipliers  $\rho_{v,\eta}^-, \rho_{v,\eta}^+$  give*

$$G_c = \sum_{(v,\eta):c(v)=c} w_{v,\eta} (\rho_{v,\eta}^+ - \rho_{v,\eta}^-), \quad H_\delta = \sum_{(v,\eta)} w_{v,\eta} \omega_{c(v)} (\rho_{v,\eta}^+ + \rho_{v,\eta}^-). \tag{24}$$

For any other feasible design  $(\tilde{u}, \tilde{\delta})$ ,

$$V(\tilde{u}, \tilde{\delta}) \leq V(u, \delta) + G^\top (\tilde{u} - u) + H_\delta (\tilde{\delta} - \delta). \quad (25)$$

This statement includes physical/corridor ties and zero-width intersections.

*Proof.* At a reached pair let  $d = r_v - q_v x - a_v s$ . Choose effective lower and upper multipliers  $\ell = \max(-d, 0)$ ,  $u^+ = \max(d, 0)$ . A positive lower multiplier occurs only when  $x = l_v^{\text{eff}}$ ; a positive upper multiplier only when  $x = h_v^{\text{eff}}$ . Among the original lower inequalities, at least one attains  $l_v^{\text{eff}}$ . Assign nonnegative portions summing to  $\ell$  only to the attaining physical and corridor inequalities. Do the analogous split for  $u^+$ . At a tie any such split works. At zero width both lower and upper normal directions are available, and the sign choice still satisfies stationarity; no division by the width is used.

All assigned original multipliers satisfy complementarity because each receiving bound is active. They therefore form full fixed-query Lagrange multipliers with the root price and reflected cap multipliers. Keep them fixed while changing only the corridor right-hand sides. A corridor upper bound contributes  $\rho^+(u_c + \delta\omega_c)$  and a lower bound contributes  $-\rho^-(u_c - \delta\omega_c)$  to the dual objective. Their change gives (24). Weak duality and exact equality at the anchor prove (25).

The certificate's old price labels merely aggregate the original history multipliers. Those multipliers can be assigned to the same public histories under any candidate policy, because the public process is exogenous. Thus the upper bound applies to every new feasible policy, not just policies sharing the old state labels.  $\square$

The oracle supplies a supergradient of the fixed-table value in the design parameters. A continuous Benders-style outer master can use these cuts and feasible-query lower bounds, but no finite exact cut count or global positive-release polynomial algorithm follows. The exact graph-sized zero-release formulation, exact fixed-query compiler, and absence of a finite-cut guarantee for the outer design are distinct results.

## 6 Computational Evidence

### 6.1 Comparisons, measurement, and audit scopes

All instances are synthetic rational service agreements. The binary comparison family uses discount and payment coefficients one, two public regimes per noninitial date, independent seeded quadratic rewards, and continuation caps above a known feasible constant-tier policy. Thus an exact root promise is available without using any optimizer. The horizon ranges from four to ten dates for the unfolded comparisons, with two seeds at each horizon. A separate public-graph grid comparison extends to twelve dates. Every input and implemented policy is retained.

We implement four explicitly defined methods. The quotient compiler merges coefficient events on the public graph. The independent tree method unfolds the same input, changes variables as in Proposition 2.1, and performs occurrence-indexed exact marginal reduction without public-vertex memoization. It uses a separate interpolation implementation, not the quotient code. The generic numerical comparison applies SciPy's sparse trust-region interior-point method to the unfolded convex quadratic problem (Virtanen et al.,

2020). The primal promise method performs backward dynamic programming directly on the public graph, restricting successor promises to a grid while allowing each current action to be the continuous residual. None of these baselines receives the quotient optimum as an input.

Each case–method pair runs in a fresh sequential worker. Three repetitions are retained with their minimum, median, and maximum. Numerical-library imports are outside timed method calls and are performed in all workers. Peak memory is an absolute process high-water mark, including the common runtime and libraries; it does not estimate allocator-level incremental memory. The comparison suite repeats construction, root inversion, certificate generation, and the inherited whole-response audit separately. In the scale suite, construction, inversion, and machine minimization are repeated three times; certificate generation and an independent single-query policy audit are each timed once and explicitly identified as diagnostics. The latter checks conditional caps, exact state flow, stationarity, complementarity, root payment, and reward without checking every affine response segment. The companion specifies all tolerances, generators, and measurement boundaries.

## 6.2 What the baselines establish after the R31 reframing

The exact tree comparison is a correctness cross-check, not an algorithmic novelty claim. All eight reported tree optima equal the quotient optima as exact rational numbers. The numerical convex solves terminate successfully; their largest absolute value difference is  $3.64 \times 10^{-7}$  and their largest primal constraint residual is below  $2.63 \times 10^{-14}$ . These residuals are not exact optimality certificates.

For the implemented occurrence-indexed code and these inputs, the tree-to-quotient timing ratio ranges from approximately 1.9 to 40.7. R31 does not use those ratios as evidence against classical laminar algorithms. A normalized memoized occurrence reduction has the same recurrence as the quotient compiler. The computational conclusion from this comparison is exact agreement between two independently implemented representations.

The public-graph promise method is the more informative structural comparator because it already exploits recombination. Its policies satisfy the exact root promise and every continuation cap under rational replay. At resolution 80, the largest absolute loss among the six reported horizon–seed pairs is below  $9.77 \times 10^{-5}$ ; one instance is already exact at resolution 40. The comparison isolates what the exact dual path contributes beyond graph compression: continuous parametric closure, exact root inversion, and a finite exact execution machine without selecting a promise mesh.

The modern parametric literature discussed in Section 2 is handled theorem by theorem rather than through a mismatched software race. Klimm and Warode (2021), for example, uses a parametric flow input and different parameterization. R31 therefore does not infer practical superiority from a generic QP timing or from an unfolded-tree count. Its computational questions are instead whether the compact compiler reproduces the exact laminar optimum, whether the whole response representation exhibits the proved output-size regimes, and whether one compiled path supports cheap repeated root queries.

### 6.3 Representation tightness, graph scale, and rational precision

Theorem 3.2 supplies an analytic worst-case family: an  $n$ -vertex chain has exactly  $2n$  global knots and  $n^2 + 2n$  stored affine response segments. This lower bound is independent of implementation and closes the one-sided complexity issue raised by the referee.

The separately retained dense-knot experiment provides an empirical counterpart. With slack continuation caps and distinct local clipping thresholds, a 511-vertex instance stores 197,369 affine segments, compared with 3,161 at 63 vertices. Coefficient lengths remain small and execution needs one writable symbol. The experiment is not used as the proof of quadratic storage; it illustrates the representation regime already established analytically.

A complementary late-renewal family retains heterogeneous binding caps at the penultimate date and recombines afterward. It reaches 2,045 public vertices and 8,164 edges. It has ten global response knots, 22,419 stored segments, and largest coefficient length 2,213 bits. The optimal protocol requires four writable symbols. Construction takes a median 2.62 seconds, while separate certificate generation and single-query audit measurements take 0.87 and 1.19 seconds. This is a long rational-horizon example with sparse cap activation, not a worst-case density test.

The broadly capped coupled family has heterogeneous payment and reward coefficients and rational transitions. Its maximum coefficient length rises from 198 to 388 to 768 bits as the horizon doubles from 16 to 32 to 64 dates; a tighter-cap 128-date instance reaches 1,501 bits. These observations are consistent with, but do not prove, the polynomial bit-height theorem. The theorem itself is algebraic and includes sorting, exact comparisons, and fraction reduction.

The response representation, the reached behavior machine, and the certificate are measured separately. A small writable alphabet does not imply a small read-only response table, and a sparse knot universe does not imply short aggregate certificate numerators. This is why R31 reports  $K_*$ ,  $C_*$ , response segments, coefficient precision, and certificate diagnostics as distinct resources.

### 6.4 Repeated root queries and the value of compiling the path

A compiled root response supports repeated exact promise inversions without rebuilding all response tables. The retained experiment performs 101 evenly spaced root queries for each late-renewal size. These timings include inversion only; emitting and certifying a new policy for each promise is an additional operation. Corollary 3.5 gives the corresponding exact complexity statement: after preprocessing, root inversion needs logarithmically many exact comparisons in the number of stored knots and a constant number of rational operations.

This repeated-query experiment is aligned with the residual parametric claim. The paper does not say that every application needs the entire path. It establishes when the complete path is compact enough to compile exactly and when that preprocessing can be amortized across root commitments or contractual design queries.

## 6.5 Verification and the retained evidence boundary

All 35 inherited exact cases reproduce their complete response curves and values, with 147,157 assertion groups in the inherited independent verifier. Forty additional small graph instances agree exactly with the independently implemented tree reduction. Two hundred and two comparisons of explicitly expanded continuation plans check the behavioral equivalence; a separate case reduces two distinct incoming prices to a single necessary symbol. Exhaustive enumeration checks the ordered frontier against 2,308 arbitrary set partitions, including unequal probabilities and nonquadratic costs. Boundary tests cover 150 feasible corridor anchors, 96 tied-normal occurrences, and 1,185 exact supporting inequalities. The test suite also rejects deliberately invalid inputs and altered certificates. These are implementation and finite-instance checks, not independent mathematical proofs of universal claims.

The earlier deployment study is neither removed nor used as favorable algorithmic evidence. At the strict tolerance, 83 of 84 cache queries refreshed and no cache speed advantage was measured. Switching paths, comparator-adjusted rents, continuous-state envelopes, correlated certificates, and governance results retain their original statements, assumptions, proofs, and numerical records in the supplementary theory volumes. The R31 response-complexity theorem and the earlier cache study address different algorithms and different questions.

## 7 Conclusions

The accepted-service problem studied here sits inside classical laminar allocation, and normalized memoization of the occurrence-indexed reduction gives the same public-graph recursion. The paper’s contribution is therefore not the existence of a scalar marginal price or memoization. It is a sharp representation theorem for the complete response on the compact graph. The quotient has only linearly many distinct knot locations, its quadratic total response-storage order is unavoidable in the worst case, and the coefficient-event compiler has polynomial exact rational complexity.

Execution is a different resource question. For a fixed root promise, the compiler generates a finite deterministic output transducer from a continuous control problem. Classical behavioral minimization yields its coarsest behavior-preserving partition. Contract structure then adds ordered price classes and a worst-case-tight linear bound on the additional writable alphabet. The public vertex, the writable symbol, and the read-only response representation are accounted for separately;  $K_*$  is not presented as total controller complexity.

For renewal agreements, the value of a limited alphabet admits an ordered allocation formula under heterogeneous probabilities and costs. Along every strictly ordered mean-preserving radial spread of continuation caps, insufficient deterministic memory has strictly increasing loss, while the exact alphabet requirement can jump at arbitrarily small spread. This is a radial comparative static, not a general convex-order claim.

These conclusions rely on a scalar additive commitment, separable strictly concave quadratic rewards, interval controls, exogenous acyclic public transitions, and public-state-sufficient caps. They do not assign the same finite-price guarantee to switching, coupled commitments, or continuous outer shared-table design. Those models and the previously established adaptation, sensitivity, and deployment results retain

their assumptions and proofs in the preserved theory volumes. The computational evidence establishes exact agreement and representation behavior for the implemented methods; it does not establish universal algorithmic dominance or empirical service calibration.

**Code, data, and reproducibility.** The R31 review package is collected in the revision directory identified in the repository README. It contains the revised theorem modules, point-by-point referee response, literature and novelty audits, derivation provenance, and a self-contained copy of unchanged shared-comparator, table, and electronic-companion text. The R30 implementation and exact audit corpus remain preserved and are the computational baseline for the unchanged algorithms. No adverse or negative result is removed.

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Table 1: Comparison on unfoldable instances

| Dates | Seed | Vertices | Occurrences | Quotient | Tree   | Convex | Value error |
|-------|------|----------|-------------|----------|--------|--------|-------------|
| 4     | 13   | 7        | 15          | 0.0009   | 0.0018 | 0.0265 | 3.64e-07    |
| 4     | 29   | 7        | 15          | 0.0009   | 0.0022 | 0.0697 | 9.12e-10    |
| 6     | 13   | 11       | 63          | 0.0023   | 0.0099 | 0.0713 | 5.88e-09    |
| 6     | 29   | 11       | 63          | 0.0022   | 0.0090 | 0.1183 | 8.74e-10    |
| 8     | 13   | 15       | 255         | 0.0040   | 0.0379 | 0.2627 | 1.36e-09    |
| 8     | 29   | 15       | 255         | 0.0039   | 0.0483 | 0.3436 | 2.57e-09    |
| 10    | 13   | 19       | 1023        | 0.0041   | 0.1663 | 1.2747 | 1.39e-08    |
| 10    | 29   | 19       | 1023        | 0.0059   | 0.1670 | 2.4538 | 3.22e-10    |

Times are seconds. Quotient time is the sum of the separately measured phase medians for construction, root inversion, and certificate generation; the exact audit is excluded. Tree and convex times are end-to-end medians of three repetitions. Tree and quotient values agree as rational numbers. Value error is the absolute difference between the numerical convex-solver value and the exact optimum; its feasibility and stationarity residuals are retained separately. All methods use the same input, not an optimum supplied by another method.

Table 2: Primal promise discretization on the public graph

| Dates | Seed | $Q = 10$ | $Q = 20$ | $Q = 40$ | $Q = 80$ | Seconds | States |
|-------|------|----------|----------|----------|----------|---------|--------|
| 4     | 13   | 8.82e-03 | 5.11e-03 | 0        | 0        | 0.0085  | 619    |
| 4     | 29   | 3.21e-02 | 7.48e-03 | 1.50e-04 | 5.29e-06 | 0.0084  | 599    |
| 8     | 13   | 2.54e-02 | 4.00e-03 | 3.79e-04 | 7.69e-05 | 0.0342  | 2275   |
| 8     | 29   | 2.19e-02 | 8.73e-04 | 2.25e-04 | 5.95e-05 | 0.0333  | 2279   |
| 12    | 13   | 1.68e-02 | 1.87e-03 | 4.15e-04 | 9.76e-05 | 0.0792  | 4971   |
| 12    | 29   | 2.15e-02 | 9.97e-04 | 3.15e-04 | 5.68e-05 | 0.0853  | 4967   |

The four resolution columns report the exact feasible-policy value loss relative to the quotient optimum. Child promises are multiples of  $1/Q$ ; local actions are continuous residuals and the root promise is not rounded. Seconds and stored states refer to  $Q = 80$ . Times are medians of three repetitions. The grid is a public-graph dynamic program, not a scenario-tree baseline; exact rational replay checks its feasibility. A zero gap on a particular instance is not a uniform exactness claim for the grid.

Table 3: Response representation and rational coefficient growth

| Family       | Vertices | Edges | Knots | Segments | Largest | Bits |
|--------------|----------|-------|-------|----------|---------|------|
| Coupled      | 61       | 228   | 128   | 2428     | 83      | 198  |
| Coupled      | 125      | 484   | 240   | 10168    | 159     | 388  |
| Coupled      | 253      | 996   | 436   | 34748    | 279     | 768  |
| Tight caps   | 509      | 2020  | 723   | 67212    | 295     | 1501 |
| Late renewal | 253      | 996   | 10    | 2706     | 11      | 277  |
| Late renewal | 509      | 2020  | 10    | 5523     | 11      | 554  |
| Late renewal | 1021     | 4068  | 10    | 11154    | 11      | 1107 |
| Late renewal | 2045     | 8164  | 10    | 22419    | 11      | 2213 |
| Dense knots  | 63       | 122   | 96    | 3161     | 68      | 12   |
| Dense knots  | 127      | 250   | 192   | 12473    | 132     | 14   |
| Dense knots  | 255      | 506   | 384   | 49529    | 260     | 16   |
| Dense knots  | 511      | 1018  | 768   | 197369   | 516     | 18   |

Knots counts the distinct global union; segments counts the entire stored family. Bits is the largest numerator or denominator bit length among reduced affine coefficients. Late-renewal caps are heterogeneous and active only at the penultimate date; this family is deliberately distinct from the dense-knot stress family and from the broadly capped coupled family. It tests long rational horizons with a small knot universe, not worst-case quadratic storage.

Table 4: Separate construction, execution, and audit costs

| Family       | Vertices | Build  | Invert | Certificate | Audit  | Symbols | Memory |
|--------------|----------|--------|--------|-------------|--------|---------|--------|
| Coupled      | 61       | 0.1220 | 16.3   | 0.0484      | 0.0775 | 18      | 97.3   |
| Coupled      | 125      | 0.5158 | 183.1  | 0.1154      | 0.2222 | 23      | 117.5  |
| Coupled      | 253      | 2.3536 | 433.9  | 0.6680      | 1.1607 | 61      | 227.2  |
| Tight caps   | 509      | 7.1123 | 1426.9 | 0.5700      | 0.1233 | 1       | 396.5  |
| Late renewal | 253      | 0.0976 | 19.1   | 0.0154      | 0.0293 | 4       | 97.2   |
| Late renewal | 509      | 0.2551 | 24.8   | 0.0483      | 0.0839 | 4       | 107.4  |
| Late renewal | 1021     | 0.7465 | 48.9   | 0.1691      | 0.2958 | 4       | 137.3  |
| Late renewal | 2045     | 2.6156 | 127.1  | 0.8716      | 1.1871 | 4       | 251.0  |
| Dense knots  | 63       | 0.0417 | 69.7   | 0.0043      | 0.0037 | 1       | 94.3   |
| Dense knots  | 127      | 0.1679 | 146.2  | 0.0163      | 0.0092 | 1       | 100.2  |
| Dense knots  | 255      | 0.6180 | 271.4  | 0.1215      | 0.0153 | 1       | 121.4  |
| Dense knots  | 511      | 2.5074 | 504.5  | 0.3282      | 0.0315 | 1       | 214.3  |

Build, certificate, and audit are seconds; inversion is microseconds. Build and inversion use three repetitions. Certificate generation and the independent single-query policy audit are each measured once in this scale suite; they are diagnostic measurements, not repeated-time claims. Symbols is the exact minimized writable alphabet. Memory is the absolute fresh-worker process high-water mark in MiB, including common Python, NumPy, and SciPy imports; it is not incremental algorithm allocation. Whole-curve audits are reported separately from these single-query KKT/flow audits.